Modeling NYC Subway Major Incidents Using Weather and Ridership

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Motivation

Every day, millions of New Yorkers rely on the subway, yet **major incidents** (delays affecting 50 or more trains) regularly disrupt service.

Two natural questions:

- How much do **weather conditions** (snow, rain, heat) drive these disruptions?
- Does **ridership** – how heavily the system is being used – explain more of the variation?

This project analyzes **monthly major incidents** from 2015–2024 and builds a statistical model combining both.

Research Question

Primary Research Question

Among months between 2020–2024 in the NYC subway system, is the **monthly count of major incidents** associated with **monthly total snowfall** and **monthly subway ridership**, and is the relationship better described as linear or multiplicative?

- **Population:** All calendar months of NYC subway operations, 2020–2024 ($n = 60$).
- **Outcome:** Monthly count of major incidents (≥ 50 trains delayed).
- **Predictors:** Total monthly snowfall (in.), total monthly ridership (millions), and secondary weather variables.
- **Summary statistics:** Means, SDs, Pearson correlations, regression coefficients, R^2, AIC.
- **Analytical method:** *Association between variables* via descriptive correlation and inferential multiple regression.

Data Sources

Three public datasets were merged to the monthly level:

- **MTA Incident Data (2015–2024)** – NY Open Data, one row per major incident with category, line, and timestamp.
- **NOAA Daily Summaries** – station `USW00094728` (NYC Central Park): `TMAX`, `TMIN`, `PRCP`, `SNOW`, `SNWD`.
- **MTA Subway Hourly Ridership (2020–2024)** – NY Open Data, boarding counts aggregated to monthly totals (subway mode only).

Operational definition. A *major incident* is any incident that delays **50 or more trains**.
Unit of analysis. One calendar month. Incidents, weather, and ridership are all aggregated to this unit and joined on the month key.

Variables & Nature

Variable	Type	Role
`total`	count (incidents/month)	outcome
`total_snow`	continuous (inches)	primary predictor
`ridership_millions`	continuous (millions)	primary predictor
`avg_tmax`, `avg_tmin`	continuous (°F)	secondary predictor
`total_prcp`	continuous (inches)	secondary predictor
`heavy_precip_days`	count (PRCP ≥ 0.5)	engineered feature
`hot_days`	count (TMAX ≥ 90°F)	engineered feature
`month`	date (year-month)	time key

All features aggregated to monthly totals or averages so each row represents one month of subway operations in NYC.

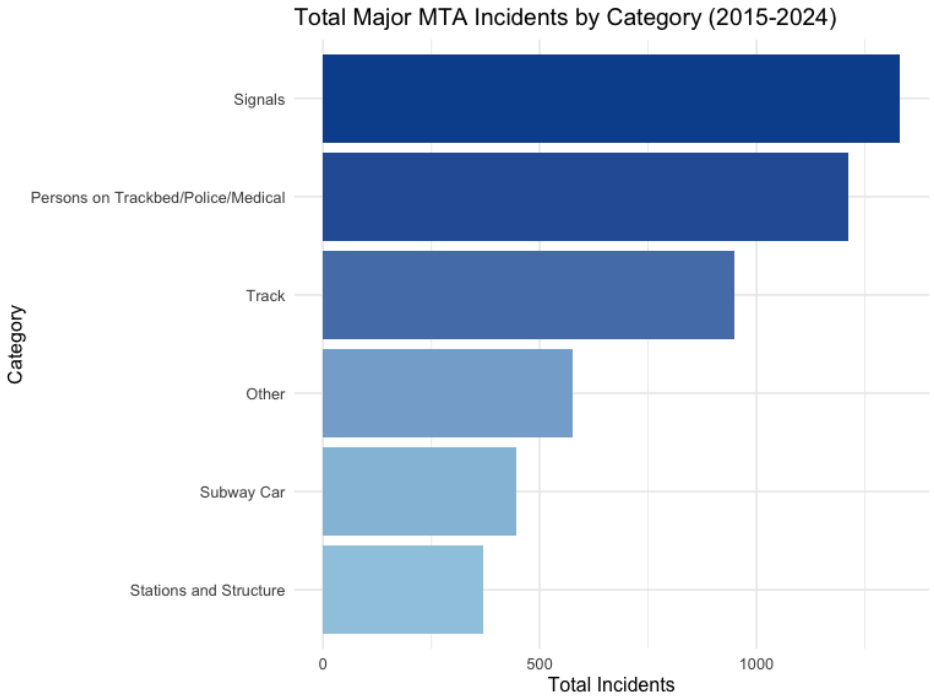
Descriptive Summary (2020–2024, $n = 60$ months)

Variable	Mean	SD	Min	Max
Major incidents / month	≈ 32	≈ 11	5	55
Total snow (inches)	≈ 2.2	≈ 5.1	0.0	25.9
Ridership (millions)	≈ 77	≈ 23	20	112
Avg. TMAX (°F)	≈ 64	≈ 16	34	89
Heavy precip days	≈ 3	≈ 2	0	9

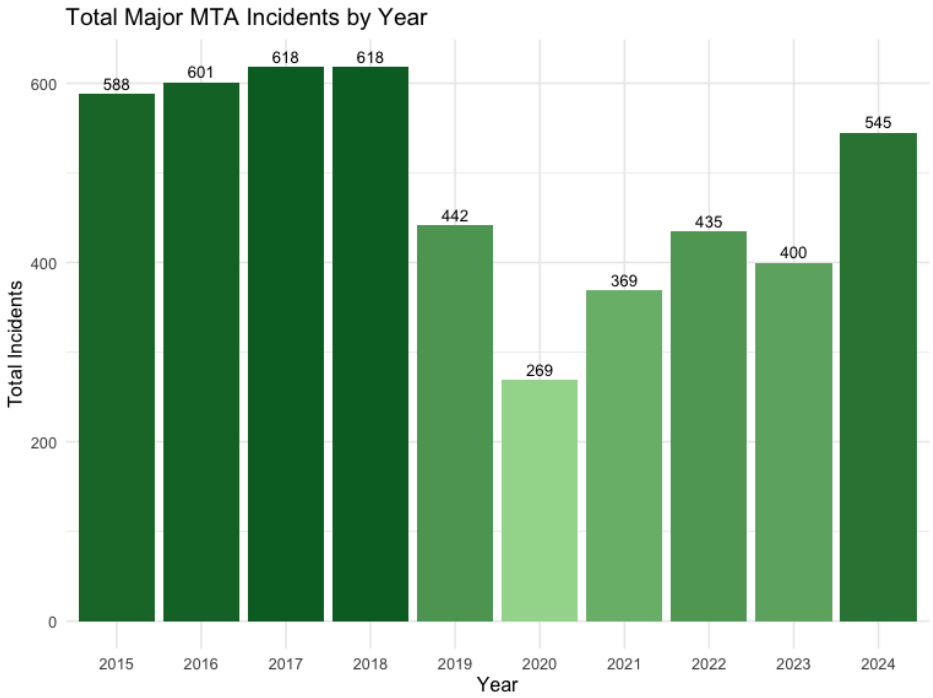
- Substantial month-to-month variation in both incidents and ridership.
- Snowfall is highly skewed – many months have zero snow, a few have large totals (hence the $\log(\text{snow} + 1)$ transformation later).

Exploratory: Incidents by Category and Year

Total Incidents by Category

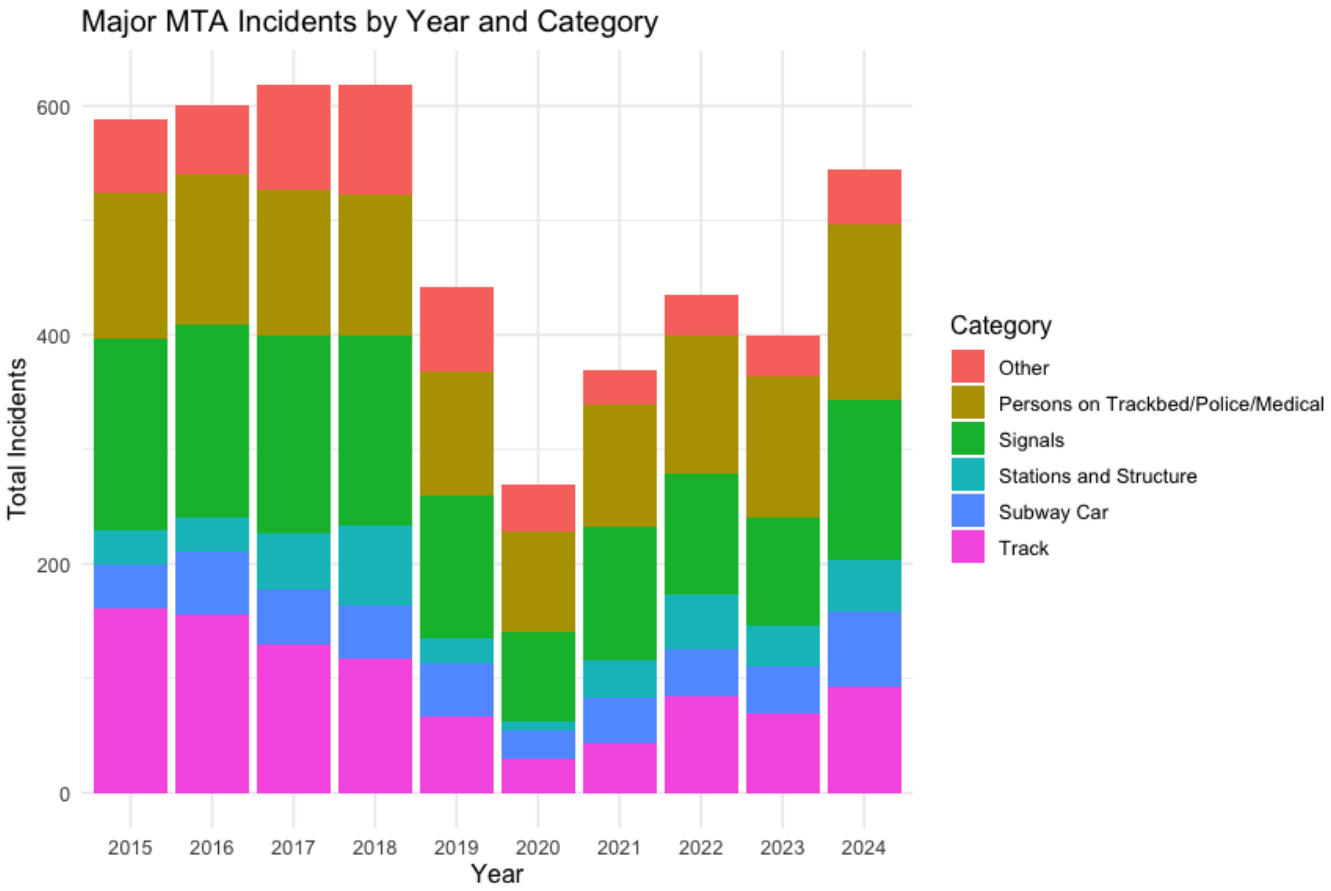


Total Incidents by Year



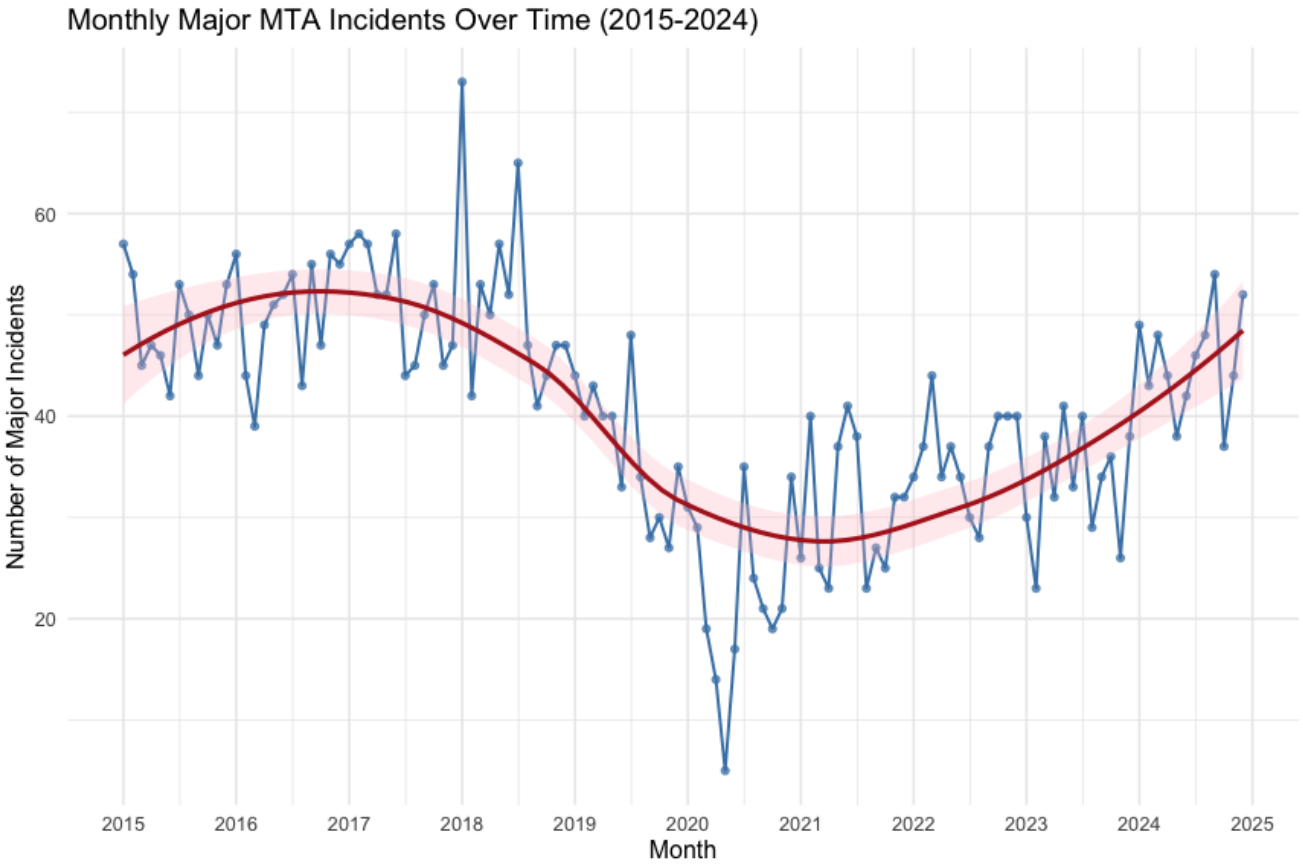
Interpretation: A handful of categories (signals, track, persons on trackbed) account for the bulk of major incidents. Annual totals show the 2020 pandemic dip and a steady recovery.

Exploratory: Year × Category Composition



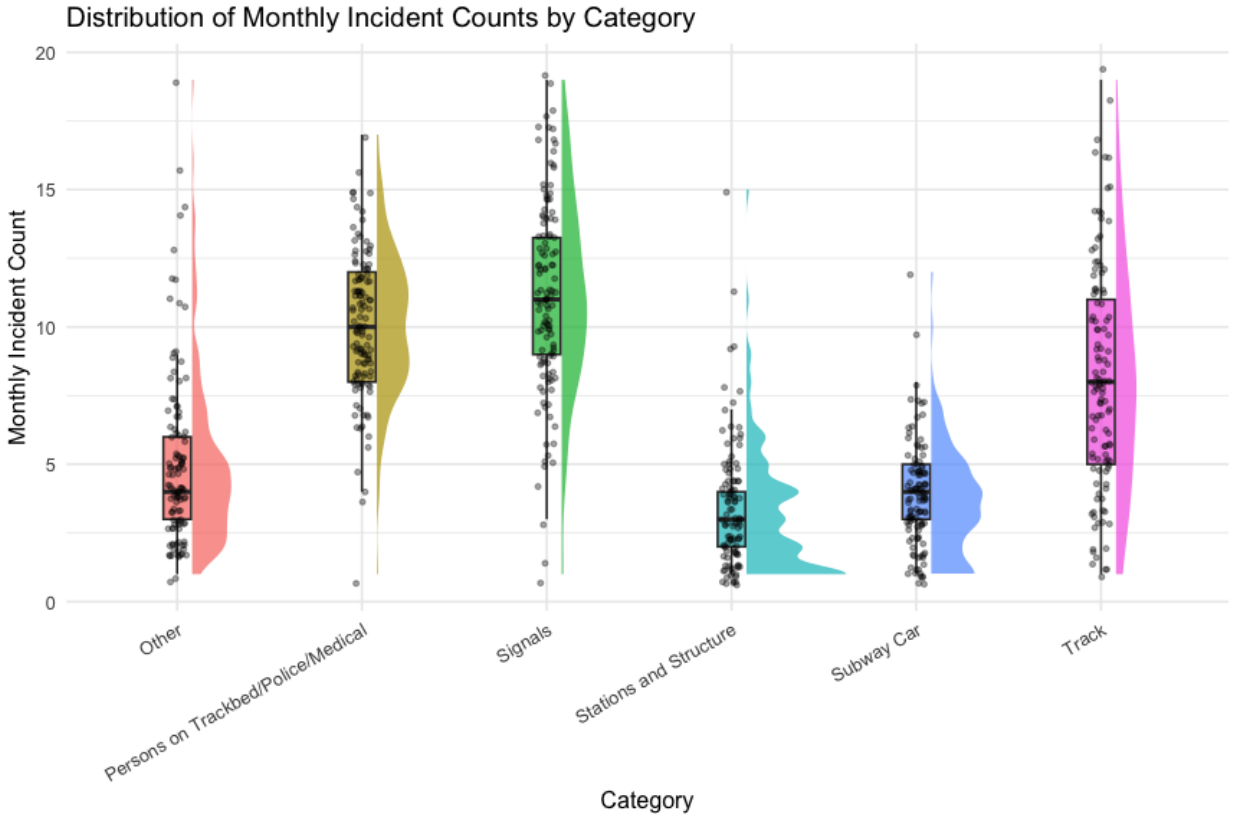
Interpretation: The stacked composition stays broadly stable across years; overall volume tracks with system usage rather than category mix shifting.

Exploratory: Monthly Incidents, 2015–2024



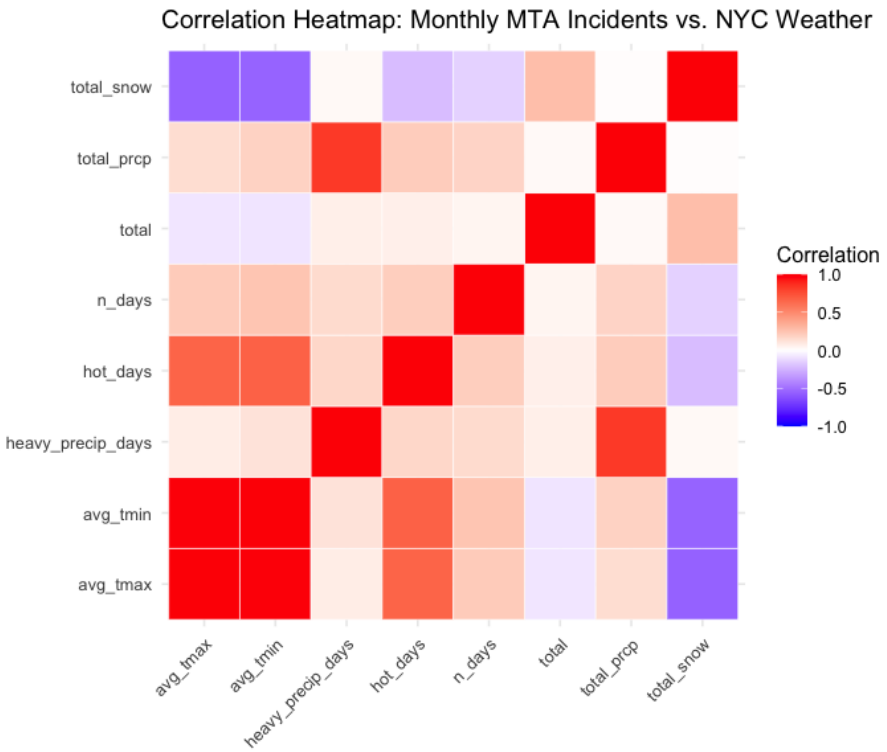
Interpretation: Across the full ten-year window, monthly incidents fluctuate seasonally; the LOESS smoother captures the 2020 ridership-driven trough and the return toward pre-pandemic levels.

Exploratory: Monthly Incidents by Category



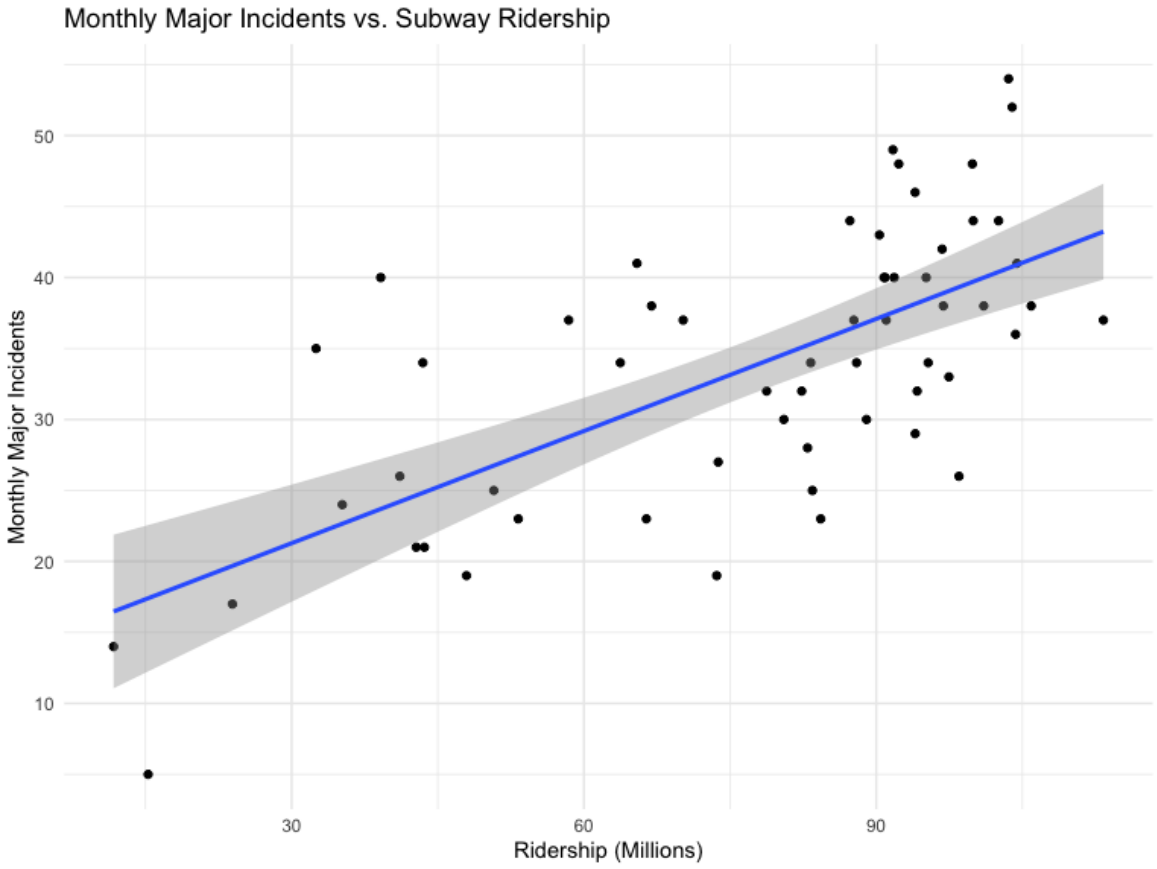
Interpretation: The half-violin pairs the density of monthly incident counts (right) with a boxplot summary (left). Categories differ in both typical level and spread; the heaviest categories occasionally produce extreme months.

Correlation Heatmap (Descriptive)



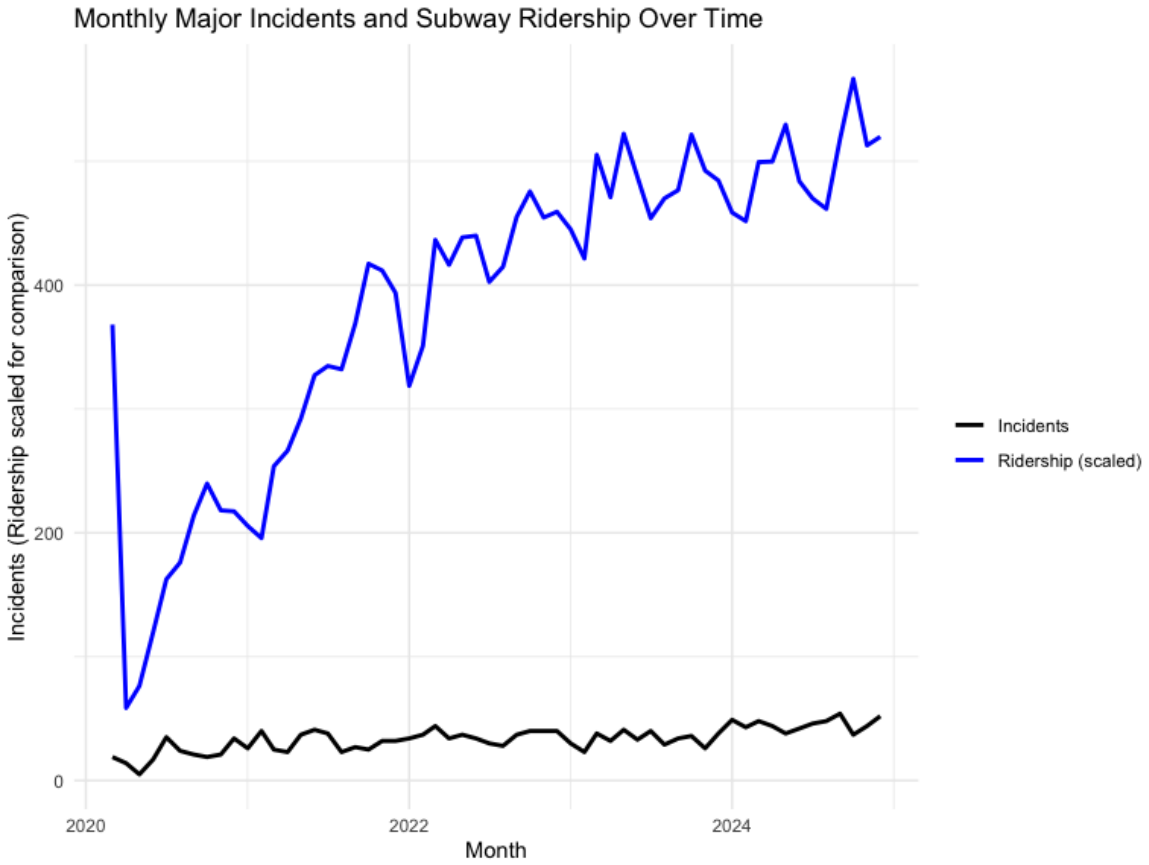
Interpretation: Among weather variables, **snowfall** has the strongest positive Pearson correlation with monthly incidents; other weather measures show weak associations. This motivates `total_snow` as the primary weather predictor.

Visualization: Ridership vs Incidents



Interpretation: Clear positive association – months with higher ridership tend to have more major incidents, suggesting *exposure* (system usage) is a key driver.

Visualization: Time Series



Interpretation: Incidents and ridership track each other over time, including the 2020 COVID-19 collapse and recovery – confirming the correlational finding at a temporal level.

Inferential Analysis: Weather-Only Models

Candidate Models

$$\text{Incidents} = \beta_0 + \beta_1 \text{avg_tmax} + \beta_2 \text{total_prcp} + \beta_3 \text{total_snow} + \varepsilon$$

$$\text{Incidents} = \beta_0 + \beta_1 \text{hot_days} + \beta_2 \text{heavy_precip_days} + \beta_3 \text{total_snow} + \varepsilon$$

Result:

- `total_snow` is statistically significant in both ($p < 0.05$).
- Other weather terms are **not** significant.
- Overall R^2 is low – weather alone is insufficient.
- VIFs all < 2; no multicollinearity concern.

Inferential Analysis: Snow + Ridership

Linear Model (2020–2024, $n = 60$)

$$\text{Incidents} = \beta_0 + \beta_1\,\text{total_snow} + \beta_2\,\text{ridership_millions} + \varepsilon$$

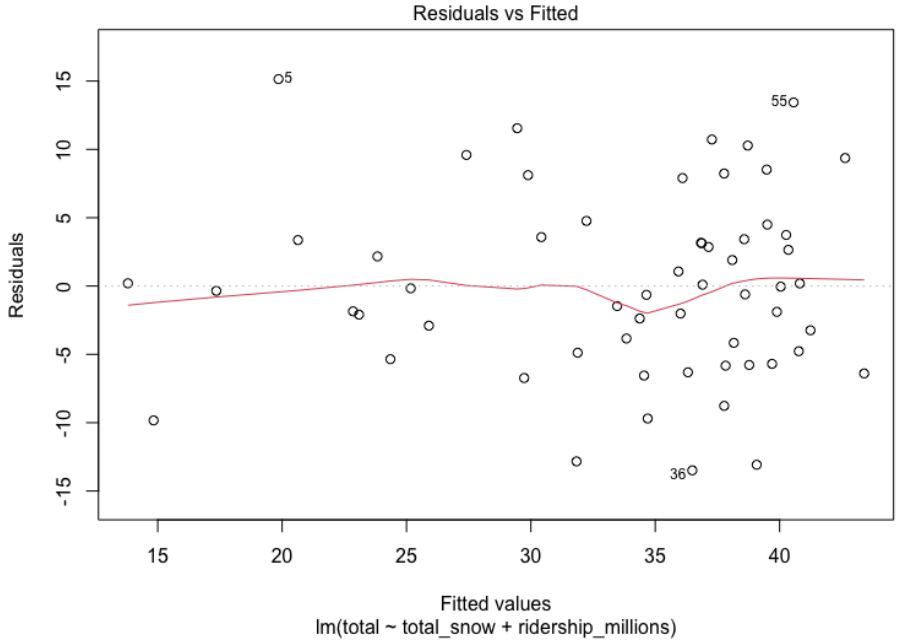
Finding. Both predictors are statistically significant ($p < 0.05$) and jointly explain a large share of variation in monthly incidents.

Interpretation.

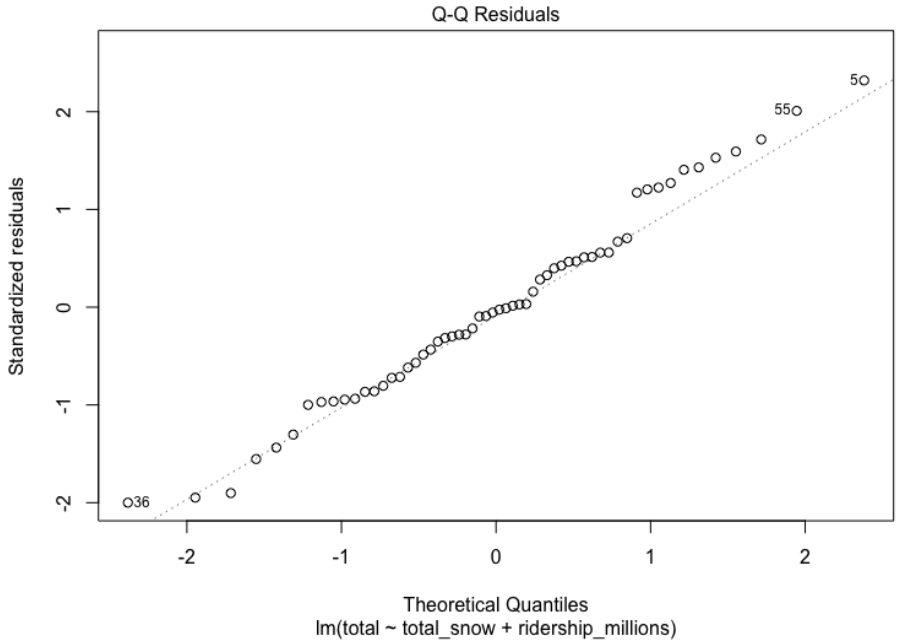
- Each additional **million riders** is associated with more incidents, holding snow constant.
- Each additional **inch of snow** is associated with more incidents, holding ridership constant.
- Large improvement in R^2 over weather-only models.

Linear Model Diagnostics

Residuals vs Fitted



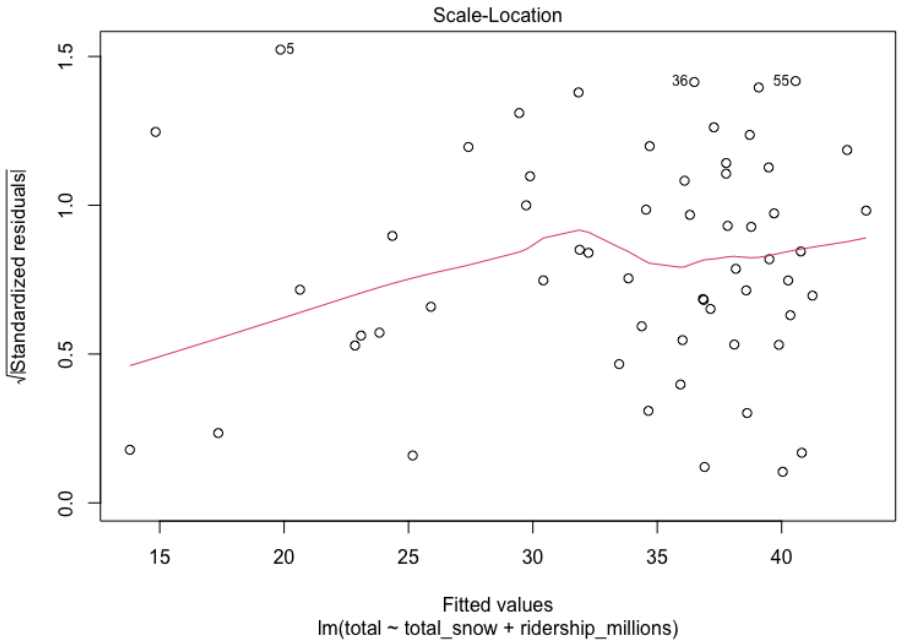
Normal Q-Q



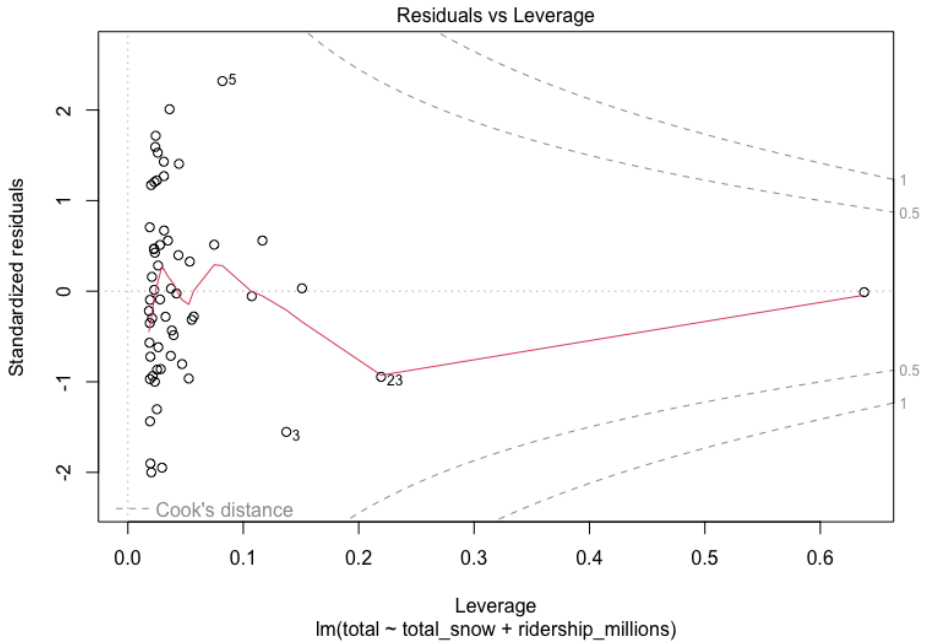
Interpretation: Residuals are roughly normal, but mild curvature in the residuals-vs-fitted plot suggests the linear form is not ideal – motivating a log transformation.

Linear Model Diagnostics (cont.)

Scale-Location



Residuals vs Leverage



Interpretation: No observations exceed Cook's distance cutoffs, but variance slightly increases with fitted values – mild heteroscedasticity.

Why a Log-Log Model?

If the true relationship between incidents, snow, and ridership is **multiplicative** (e.g., "a 10% jump in ridership produces a 5% jump in incidents"), a linear model will systematically mis-fit.

Three candidate specifications compared:

- **Linear:** $y = \beta_0 + \beta_1 s + \beta_2 r + \varepsilon$
- **Log response:** $\log(y) = \beta_0 + \beta_1 s + \beta_2 r + \varepsilon$
- **Log-log:** $\log(y) = \beta_0 + \beta_1 \log(s+1) + \beta_2 \log(r) + \varepsilon$

In the log-log model, coefficients are **elasticities**: a 1% change in the predictor is associated with a β% change in incidents.

Model Comparison via AIC

Result

The **log-log model has the lowest AIC** of the three candidates.

Model	df	AIC
Linear	4	392.13
Log response	4	12.52
Log-log	**4**	**5.40**

Interpretation: The linear model's AIC is nearly 387 units higher than the log-log – overwhelming evidence in favor of the log-log specification (any ΔAIC > 10 is considered decisive). The log response alone already improves enormously; adding log-transformed predictors gives the best fit of the three.

Log-Log Regression Coefficients

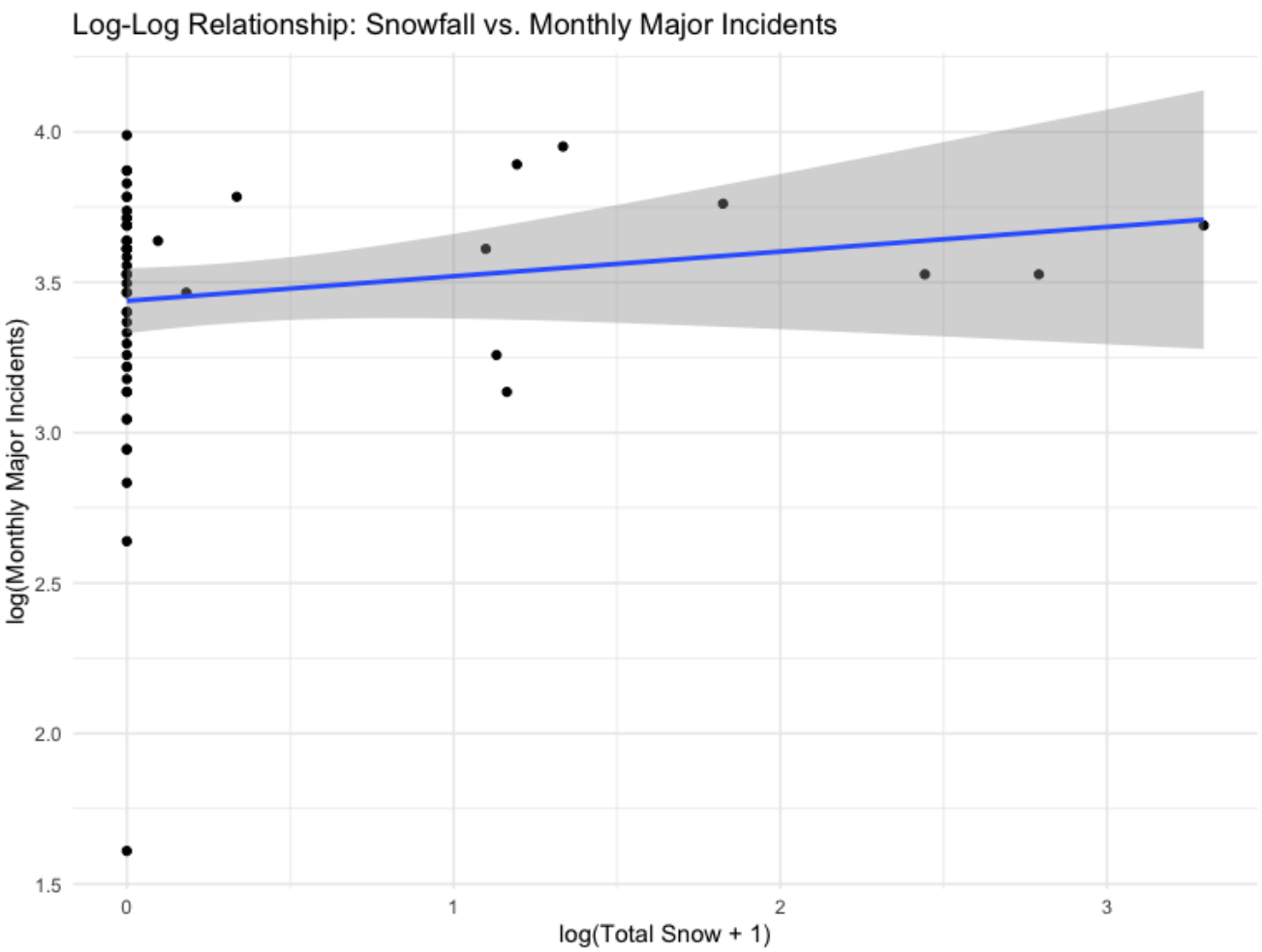
Term	Est.	SE	t	p-value	
(Intercept)	0.751	0.297	2.53	0.014	*
log(total_snow + 1)	0.135	0.045	3.03	0.004	**
log(ridership_millions)	0.626	0.069	9.12	$< 10^{-11}$	***

Residual SE $= 0.243$ on 55 df.

Elasticity interpretation:

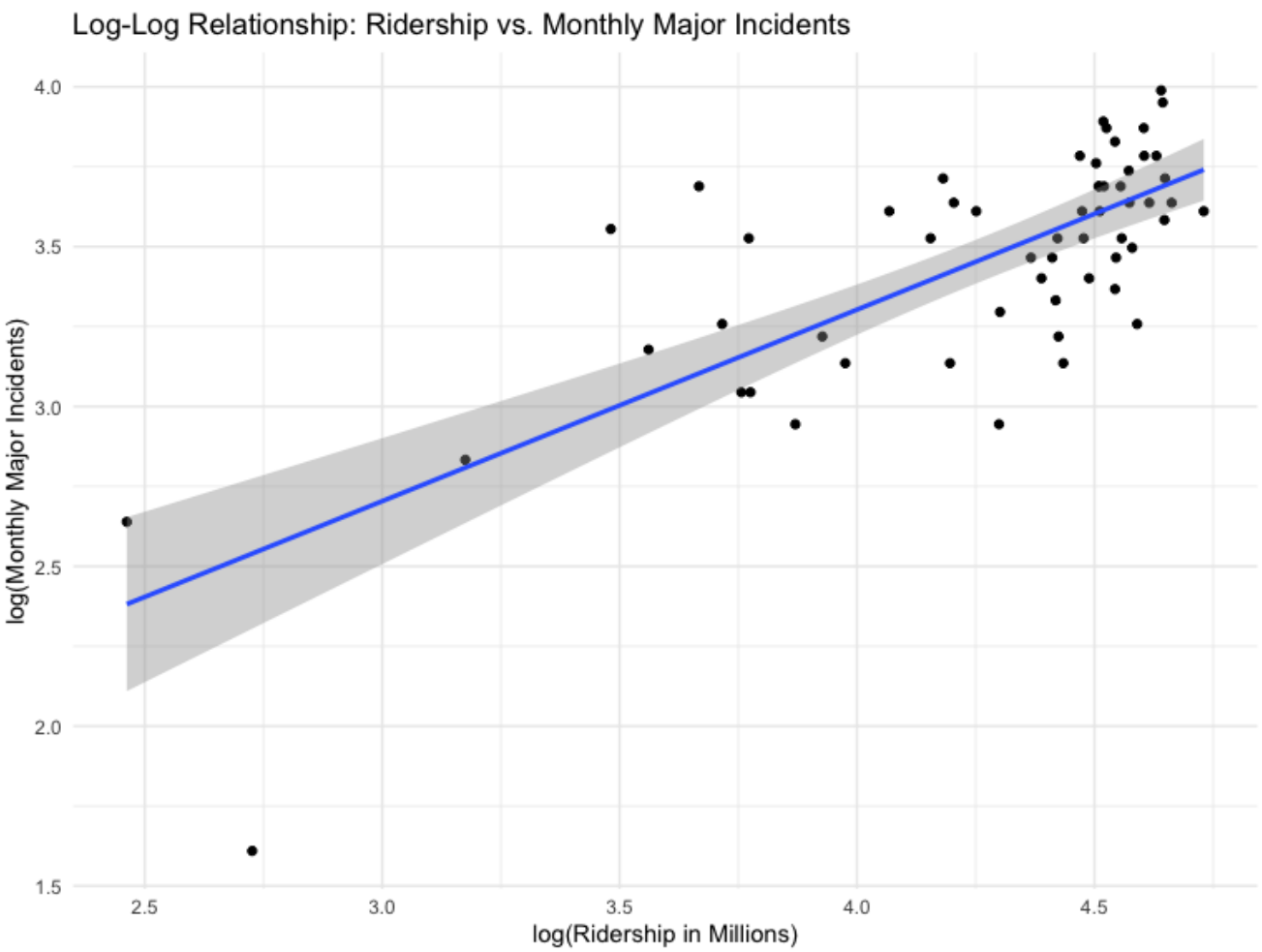
- A **1% increase in ridership** is associated with a $\approx 0.63\%$ increase in monthly major incidents.
- A **1% increase in snowfall** (plus one) is associated with a $\approx 0.13\%$ increase in incidents.
- Both coefficients are significant; ridership carries far more of the signal.

Log-Log: Snowfall vs Incidents



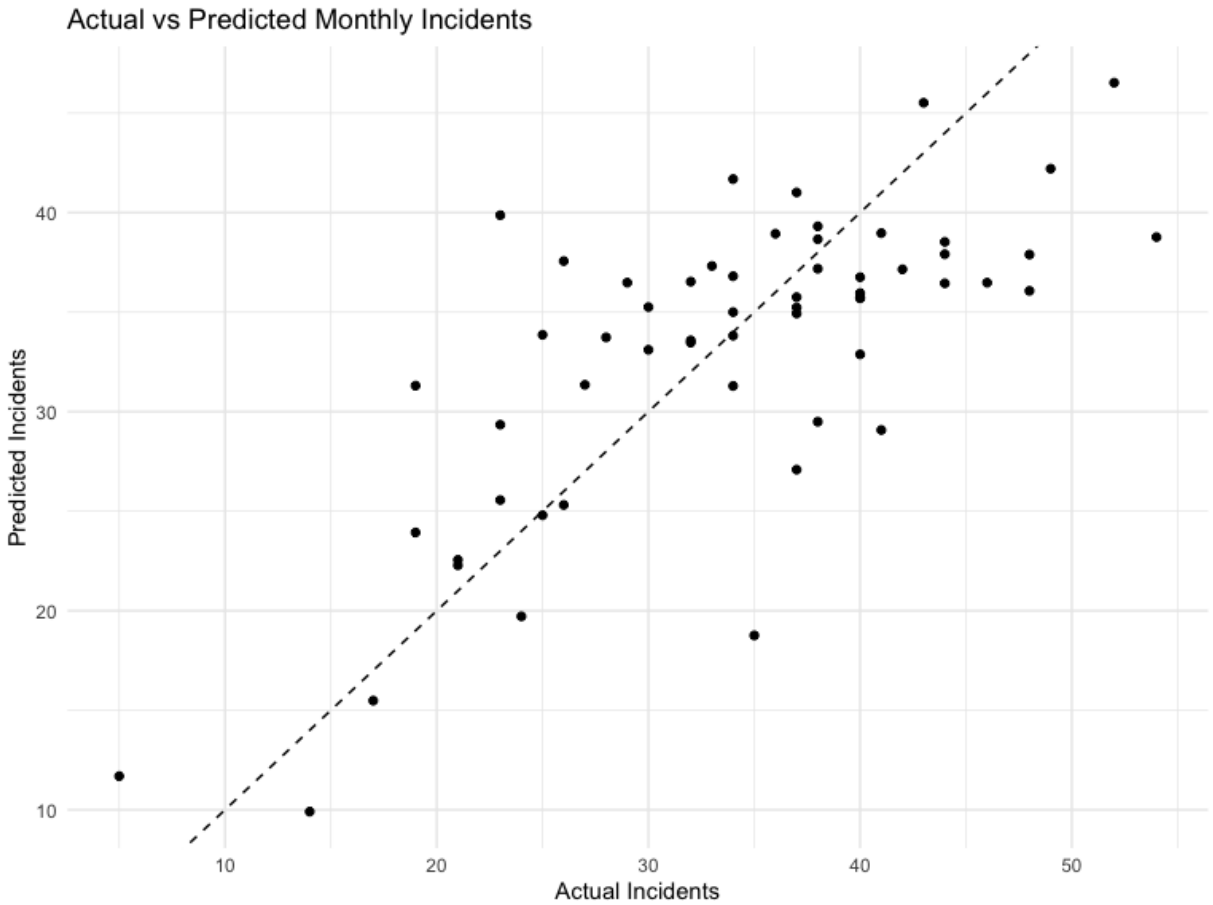
On the log scale, the snowfall–incident relationship looks visibly more linear, with a tighter spread around the fit line.

Log-Log: Ridership vs Incidents



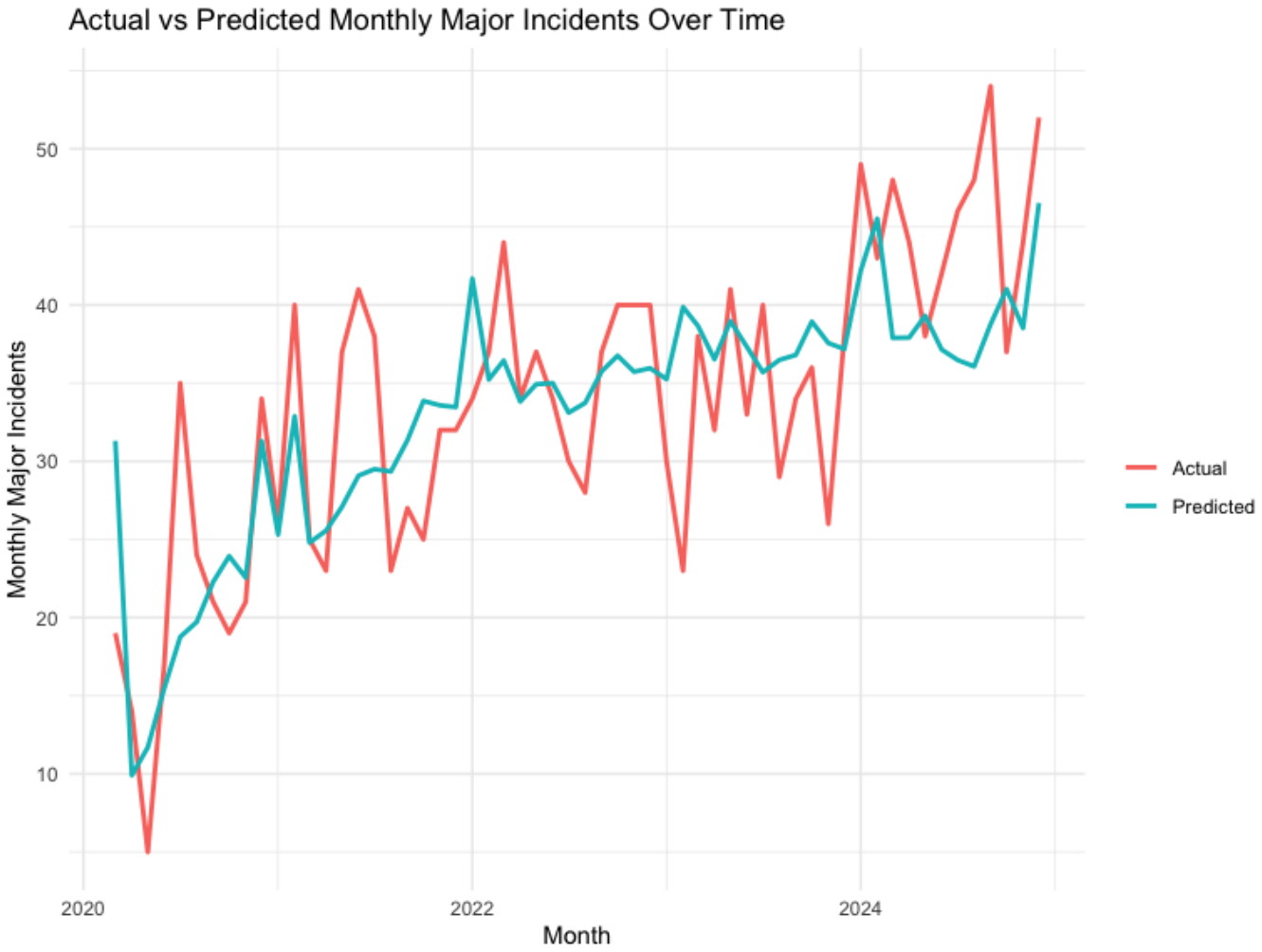
Ridership also shows a cleaner linear pattern after log transformation, consistent with an elasticity-style interpretation.

Actual vs Predicted Incidents



Points cluster around the 45° reference line: the log-log model's predictions track observed monthly incident counts well.

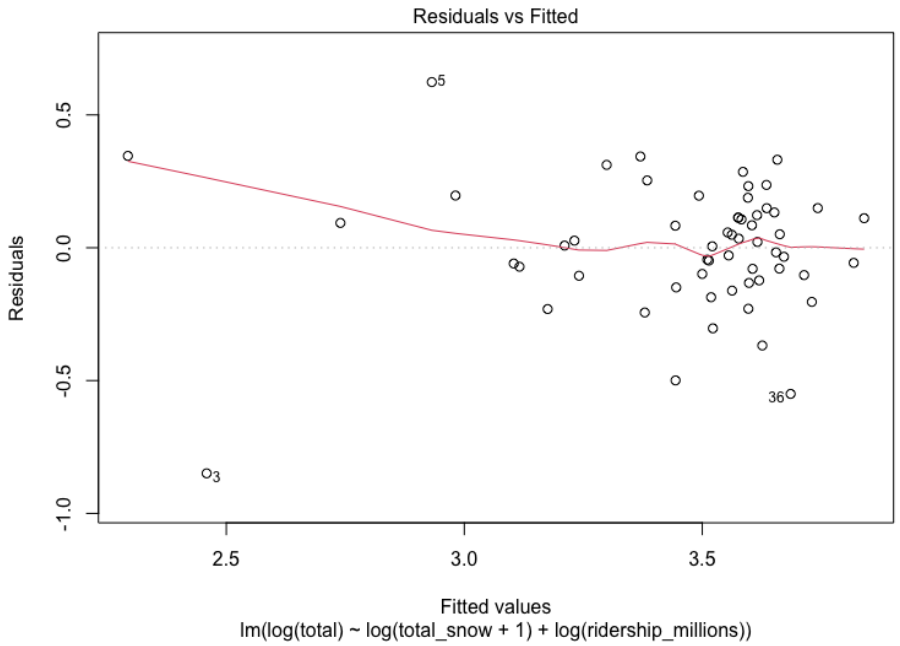
Actual vs Predicted Over Time



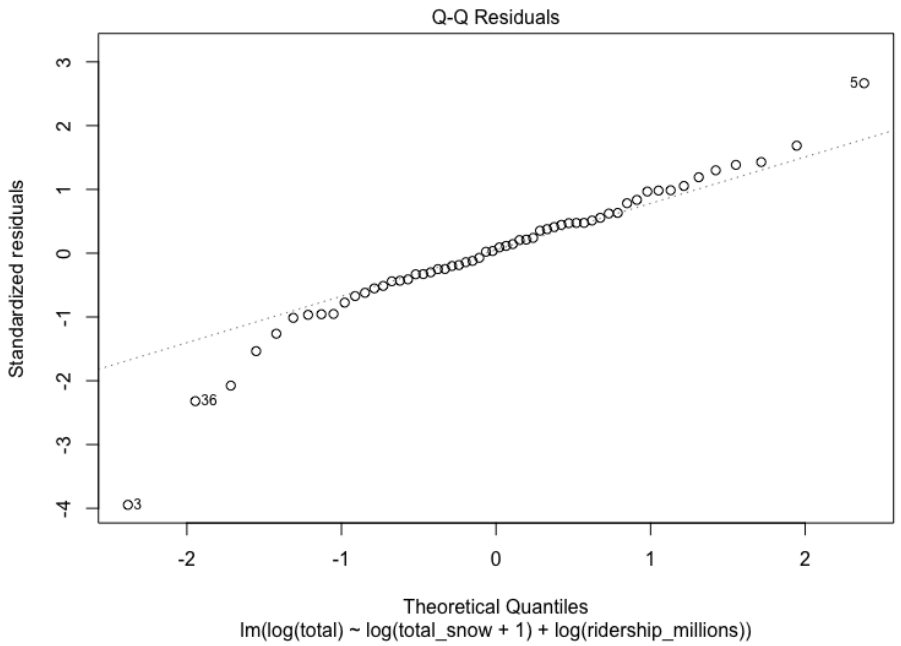
Predicted series closely follows the observed monthly incident totals, including the 2020 collapse and 2021–2024 recovery.

Log-Log Model Diagnostics

Residuals vs Fitted



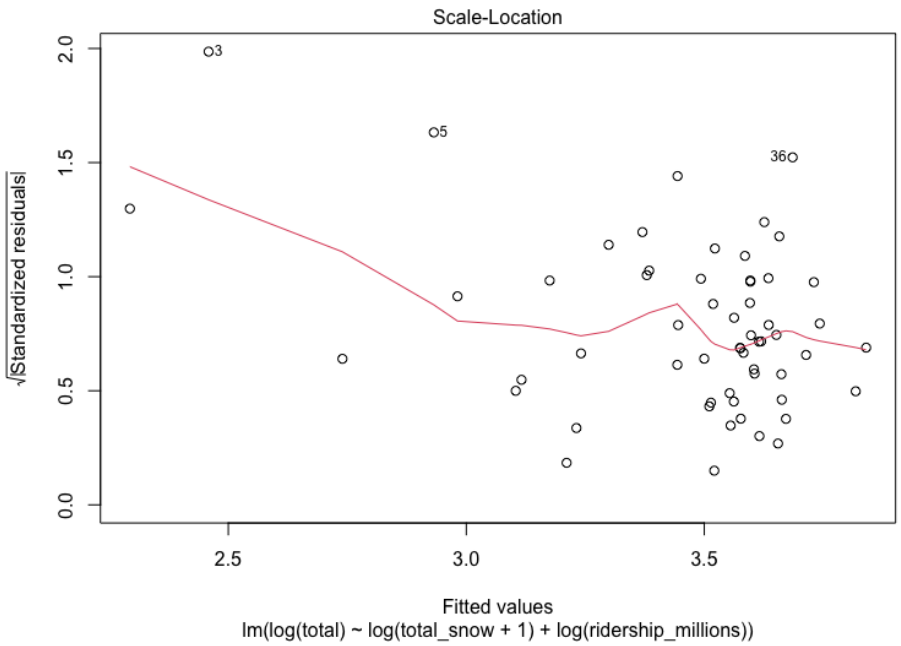
Normal Q-Q



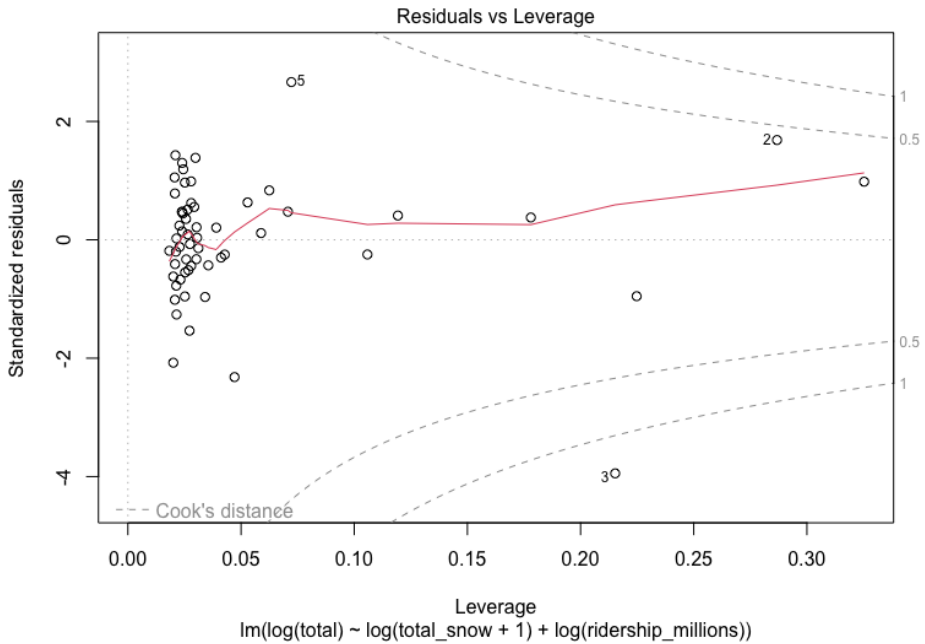
Interpretation: Residuals behave more like iid normal than in the linear model – tighter, more symmetric, less curvature.

Log-Log Model Diagnostics (cont.)

Scale-Location



Residuals vs Leverage



Interpretation: Variance is noticeably more stable across fitted values, and no points exceed Cook's-distance cutoffs. The log-log specification passes diagnostics more cleanly.

Answering the Research Question

Research Question (restated)

Is the monthly count of NYC subway major incidents associated with snowfall and ridership, and is the relationship linear or multiplicative?

Answer:

- **Yes** – both **snowfall** and **ridership** are statistically significant predictors of monthly major incidents.
- **Ridership** contributes substantial explanatory power that weather alone cannot; snowfall retains an *independent* effect after controlling for ridership.
- The relationship is better described as **multiplicative**: the log-log model has the lowest AIC and the cleanest residual diagnostics.

Limitations

- **Ridership coverage** – daily ridership is only publicly available from 2020 onward, restricting the combined model to $n = 60$ months.
- **Confounding by COVID-19** – the early 2020 collapse in both ridership *and* incidents is a dominant signal; relationships may differ in steady-state periods.
- **Aggregation** – monthly totals smooth over storm-day dynamics; a single 20-inch snowstorm and a month of steady light snow look the same in `total_snow`.
- **Single weather station** – Central Park measurements may not represent borough-level conditions.
- **Association, not causation** – observational data cannot rule out unmeasured confounders (e.g., capital maintenance cycles, labor actions).

Future Work

- Break incidents down by **category** (signals, track, persons on trackbed) and model each separately.
- Add **lagged** weather effects – does a storm on day t cause incidents on day $t + 1$?
- Include **line-level** ridership instead of system-wide totals.
- Try non-linear models (GAMs, random forests) to see whether additional structure remains.
- Extend the pre-2020 period once historical ridership becomes available.

References & Code

Data:

1. MTA. *Subway Major Incidents 2015–2019* and *2020–2024*. New York State Open Data.
2. MTA. *Subway Hourly Ridership (2020+)*. New York State Open Data.
3. NOAA NCEI. *Daily Summaries* – station `USW00094728` (NYC Central Park). `https://www.ncei.noaa.gov/access/services/data/v1`

Code: `FinalVizProject.qmd` (Quarto / R).
Tools: R – `dplyr`, `ggplot2`, `readr`, `lubridate`, `tidyr`, `httr`, `car`, `ggdist`.

Thank you! Any questions?